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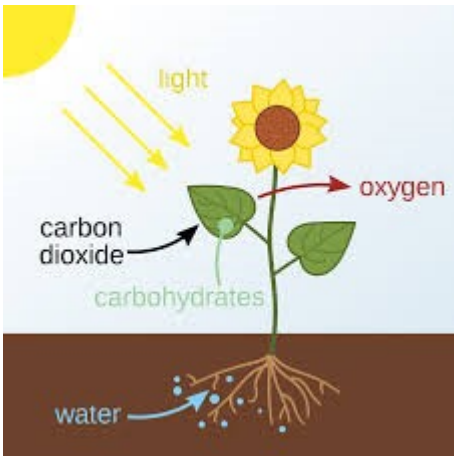
**Class 11 Biology Notes Chapter 13 Photosynthesis In Higher Plants**

## Photosynthesis

Photosynthesis is an enzyme regulated anabolic process of manufacture of organic compounds inside the chlorophyll containing cells from carbon dioxide and water with the help of sunlight as a source of energy.

$$6CO_2 + 12H_2O \xrightarrow[Chlorophyll\ +\ enzymes]{Sun\ light} C_6H_{12}O_6 + 6H_2O + 6O_2 \uparrow$$




**Historical Perspective**

**Josheph Priestley (1770):** Showed that plants have the ability to take up $CO_2$ from atmosphere and release $O_2$. (Candle with bell jar and mouse expt.)

**Jan Ingenhousz (1779):** Release of O, by plants was possible only in sunlight and only by the green parts of plants. (Expt. with aquatic plant in light & dark)

**Theodore de Saussure (1804):** Water is an essential requirement for photosynthesis to Occur.

**Julius Von Sachs (1854):** Green parts in plant produce glucose which is stored as starch.

**T.W. Engelmann (1888):** The effect of different wavelength of light on photosynthesis and plotted the first action spectrum of photosynthesis.

**C.B. Van Niel (1931):** Photosynthesis is essentially a light dependent reaction in which hydrogen from an oxidizable compound reduces $CO_2$ to form sugar. He gave a simplified chemical equation of photosynthesis.

$$2H_2A + CO_2 \xrightarrow{\text{Sun light}} 2A + CH_2O + H_2O$$

**Hill (1937):** Evolution of oxygen occurs in light reaction.

**Calvin (1954-55):** Traced the pathway of carbon fixation.

**Site for photosynthesis**

Photosynthesis takes place only in green parts of the plant, mostly in leaves. Within a leaf, photosynthesis occurs in mesophyll cells which contain the chloroplasts. Chloroplasts are the actual sites for photosynthesis. The thylakoids in chloroplast contain most of pigments required for capturing solar.

Energy to initiate photosynthesis: The membrane system (grana) is responsible for trapping the light energy and for the synthesis of ATP and NADPH. Biosynthetic phase (dark reaction) is carried in stroma.

## Importance of Photosynthesis

- Synthesis of organic compounds.
- Change of radiant energy into chemical energy.
- Useful products are obtained from plants gums, oils timber fire wood, resins rubber, fibers and drugs, etc.
- Balance the percentage of $O_2$ and $CO_2$ in atmosphere.
- Fossil fuels like coal, natural gas and petroleum have been formed inside the earth indirectly as a product of photosynthesis.

## Pigments involved in photosynthesis

**Chlorophyll a:** (Bright or blue green in chromatograph). Major pigment, act as reaction center, involved in trapping and converting light into chemical energy. It is called universal photo-synthetic pigment.

**Chlorophyll b:** (Yellow green)

**Xanthophyll's:** (Yellow)

**Carotenoids:** (Yellow to yellow-orange)

In the blue and red regions of spectrum shows higher rate of photosynthesis.

**Light Harvesting Complexes (LHC)**

The light harvesting complexes are made up of hundreds of pigment molecules bound to protein within the photosystem I (PS-1) and photosystem II (PS-II). Each photosystem has all the pigments except one molecule of chlorophyll 'a' forming a light harvesting system (antennae). The reaction center (chlorophyll a) is different in both the photosystems.

**Photosystem I (PS-I):** Chlorophyll 'a' has an absorption peak at 700 nm (P700).
**Photosystem II (PS-11):** Chlorophyll 'a' has absorption peak at 680 nm (P680),

**Process of photosynthesis.**

It includes two phases-Photochemical phase and biosynthetic phase. (Formerly known as Light reaction and dark reaction)

**Photochemical phase (Light reaction):** This phase includes-light absorption, splitting of
water, oxygen release and formation of ATP and NADPH. It occurs in grana region of chloroplast.

**Biosynthetic phase (Dark reaction):** It is light independent phase, synthesis of food material (sugars). (Calvin cycle). It occurs in stroma region of chloroplast.

**Photophosphorylation**

The process of formation of high-energy chemicals (ATP and NADPH) in presence of light.

**Non-Cyclic photophosphorylation**

Two photosystems work in series First PSII and then PSI. These two photosystems are connected through an electron transport chain (Z. Scheme). Both ATP and NADPH + H+ are synthesized by this process. PSI and PSII are found in lamellae of grana, hence this process is carried here.

## Cyclic photophosphorylation

Only PS-I works, the electron circulates within the photosystem. It happens in the stroma lamellae (possible location) because in this region PSII and NADP reductase enzyme are absent. Hence only ATP molecules are synthesized. It occurs when only light of wavelengths beyond 680 nm are available for excitation.

## The Electron Transport System

1. Reaction center of photosystem II absorbs light of 680 nm in red region and causing electron to become excited. These electrons are picked by an electron acceptor which passes to electron transport system consisting of cytochromes.
2. Electrons are passed down the electron transport chain and then to the pigment of PSI.
3. Electron in the PSI also get excited due to light of wavelength 700nm and are transferred to another accepter molecule having a greater redox potential.
4. When electron passes in downhill direction, energy is released. This is used to reduce the ADP to ATP and NADP+ to NADPH. The whole scheme of transfer of electron is called Z-scheme due to its shape.
5. Photolysis of water release electrons that provide electron to PS II. Oxygen is also released during this process.

**Calvin Cycle/ $C_3$ cycle/ Reductive Pentose Sugar Phosphate Pathway**

Malvin Calvin, Benson and their colleagues used radioactive 14C and Chlorella and Scenedesmus algae to discover that first $CO_2$ fixation product is 3-carbon organic compound (3-phosphoglyceric acid) or PGA. Later on a new compound was discovered which contain 4-carbon called Oxaloacetic Acid (AAO). On the basis of number of carbon atoms in first stable product they are named C3 and C pathway.

**Calvin cycle can be described under three stages: carboxylation, reduction and regeneration.**

1. Carboxylation is the fixation of into 3-phosphoglyceric acid (3-PGA). Carboxylation of RuBP occurs in presence of enzyme RuBP carboxylase (RuBisCO) which results in the formation of two molecules of 3-PGA.
2. Reduction is series of reaction that leads to formation of glucose. Two molecules of ATP and two molecules of NADPH are required for reduction of one molecule of. Six turns of this cycle are required for removal of one molecule of Glucose molecules from pathway.
3. Regeneration is the generation of RuBP molecules for the continuation of cycle. This process requires one molecules of ATP.
4. For every molecule of entering the Calvin Cycle, 3 molecules of ATP and 2 molecules of NADPH is required. To make one molecules of glucose 6 turns of cycle is completed so total energy molecule required is.

| In | Out |
|---|---|
| Six $CO_2$ | One glucose |
| 18 ATP | 18 ADP |
| 12 NADPH | 12 NADP |

## $C_4$ pathway/ Hatch Slack Pathway

● This pathway was worked out by Hatch and Slack (1965, 1967), mainly operational in plants growing in dry tropical region like Maize, Sugarcane, Sorghum etc.

● In this pathway first stable product is a 4-carbon compound Oxaloacetic acid (AAO) so called as $C_4$ pathway. $C_4$ plants have Kranz Anatomy (vascular bundles are surrounded by bundle sheath cells arranged in wreath like manner), characterized by large no of chloroplast, thick wall impervious to gases and absence of intercellular spaces.

● The primary $CO_2$ acceptor is a 3-carbon molecule Phosphoenol Pyruvate present in mesophyll cells and enzyme involved is PEP carboxylase.

● OAA formed in mesophyll cell forms 4-carbon compound like malic acid or aspartic acid which is transported to bundle sheath cells.

● In bundle sheath cell, it is broken into $CO_2$ and a 3-carbon molecule. The 3-carbon molecule is returned back to mesophyll cells to form PEP.

● The $CO_2$ molecules released in bundle sheath cells enters the Calvin cycle, where enzyme RuBisCO is present that forms sugar.

## Photorespiration

- It is a the light dependent process of oxygenation of RuBP and release of carbon dioxide by photosynthetic organs of plants.
- Photorespiration decreases the rate of photosynthesis when oxygen concentration is increased from 2-3% to 21%.
- Presence of light and higher concentration of Oxygen results in the binding of RuBisCO enzyme with O2 to form.
- RuBis CO + $O_2$ PGA + phosphoglycolate
- This pathway involves Chloroplast, Peroxisome and Mitochondria Photorespiration do not occurs in $C_4$ plants.

## Difference between $C_3$ Plants and $C_4$ Plants

| $C_3$ Plants | $C_4$ Plants |
|---|---|
| The leaves do not have Kranz anatomy. | The leaves show Kranz anatomy in leaves. |
| Photorespiration occurs. | Photorespiration does not occur. |
| RuBisCO is the first acceptor of $CO_2$. | PEP is the first acceptor of $CO_2$ |
| PGA is the first stable product. | OAA is the first stable product. |
| Plants are adapted to all climates. | Plants are adapted to tropical climate. |
| Mesophyll cells perform complete photosynthesis. | Mesophyll cells perform only initial fixation. |

## Factors affecting photosynthesis

- Light: As light intensity increases, the rate of photosynthesis also increases until light saturation point.
- Carbon dioxide concentration: With increase in concentration of $CO_2$ rate of photosynthesis increase till the compensation point.
- Temperature: It does not influence the rate of photosynthesis directly but at higher temperature enzyme activity is inhibited due to denaturation of enzymes which affect the dark reaction.
- Water: due to increase in amount of water, rate of photosynthesis does not increase proportionally as after saturation no more water is required during photosynthesis.

**Blackman's Law of Limiting Factors states**

If a chemical process is affected by more than one factor, then its rate will be determined by the factor which is nearest to its minimal value it is the factor which directly affects the process if its quantity is changed.

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